

Deep Learning for Time Series

Details on the project

Romain Tavenard

- Task: Time Series Classification (Dataset: LSST)
- Choose one of the following settings
 - Setting 1: Adapt a foundation model
 - Setting 2: Pre-train on forecasting (Informer datasets only), adapt on classification

Project

- Requirements
 - A group = 3 or 4 students
 - A strong competitor + a decent baseline (without pre-training)
 - A report is due on **March 16th, 23:59 Paris time**: max 3 pages ICML 2026 style (incl. a link to a git repo for the code)
 - Defense: 10 minutes presentation + 5 minutes question, on the **23rd of March**
 - Each student should attend the session her/his project is assigned (morning or afternoon)

Data loading

- You can use `tslearn` to load the LSST dataset as numpy arrays:

```
from tslearn.datasets import UCR_UEA_datasets
```

```
# Load the LSST dataset from UEA archive
```

```
ds = UCR_UEA_datasets()
```

```
X_train, y_train, X_test, y_test = ds.load_dataset("LSST")
```

Declaring groups

- Upload a CSV (**one per group**, no double posting please) on the course page with columns
 - STUDENT_ID
 - STUDENT_NAME
 - STUDENT_FIRSTNAME
 - AVAILABLE_2303_MORNING
 - AVAILABLE_2303_AFTERNOON
- Deadline: **February 15th, 23:59 Paris time**